



SwarmOps automates the hardest part of operating a drone fleet: deciding, in real time, which drone flies which mission, in what order, along which route — while guaranteeing no drone runs out of battery mid-flight. A working distributed system, deployed on AWS and demonstrable live today.

VRPTW solver

Hungarian assignment + OR-Tools routing, battery-feasibility proven per route

Seconds to re-plan

Event-driven re-planning from live telemetry, no human trigger

8 microservices

On AWS EKS; all infrastructure as code (Terraform)

GitOps delivery

CI/CD + Argo CD, canary rollouts, full observability

THE PROBLEM

Beyond a handful of drones, fleet coordination is computationally intractable for a human dispatcher: battery limits driven by distance *and* payload, mission priorities and deadlines, no-fly zones, charging logistics, and mid-mission disruptions — all interacting at once. The result: wasted flight time, missed high-priority deadlines, stranded aircraft, and no single operational picture. Manufacturer software flies one drone well; nobody answers the fleet-level question — *who should do what, and can they actually complete it?*

THE PRODUCT — BUILT, NOT PLANNED

- **Optimal assignment & routing** — benchmarked against a greedy baseline; battery drain simulated per route, charging stops inserted automatically.
- **Real-time re-planning** — telemetry streams through a message bus; disruptions re-solve the affected schedule in seconds.
- **Live control room** — real-time map, mission board, fleet inventory, operational dashboards.
- **Production-grade engineering** — automated, auditable, canary-gated deployments; the maturity startups usually retrofit in year two already exists.

Current scope, stated plainly: the fleet is simulated — a deliberate capstone decision. Drone interfaces are already abstracted; real-aircraft integration (MAVLink) is Phase 1 engineering, not research risk.

MARKET

SEGMENT	USE CASE
Defense & security	Resupply tasking, patrol/surveillance coverage, inspection rounds
Infrastructure & energy	Power lines, pipelines, solar — recurring route-heavy missions
Agriculture	Survey & spraying coordination at scale
Delivery	Last-mile / site-to-site (regulation-gated)

Positioning: vendor-neutral coordination layer — the operations brain above whatever aircraft an organization already owns. Fleets are scaling, BVLOS regulation is opening, and Israel's drone ecosystem is a natural first market.

ROADMAP — 12 MONTHS

0–3 MO

Real aircraft

Incorporate; MAVLink integration; full loop on real hardware; CAA groundwork.

3–6 MO

First pilot

Design partner (inspection / site security); success criteria set up front.

6–12 MO

Productize

Multi-tenancy, pricing; pilot → paying reference; seed round.

TEAM

Tony [surname] (team lead) · **Guy [surname]** · **Harel**

[surname] — three engineers who designed, built, and operate the entire system: algorithms, services, frontend, cloud infrastructure, and delivery pipeline.

THE ASK

Incubation (workspace + program affiliation) · **Mentorship** (business development, defense/aerospace) · **Introductions** (operators, integrators, infrastructure companies) · **Pre-seed support** ([amount]: development drones, cloud, legal/regulatory). Live demonstration available on request.